

Boolean Information Retrieval System Evaluation

Measuring Precision and Recall

Introduction

Information Retrieval (IR) systems help users find relevant content in large document collections. Among the oldest and simplest IR models is the Boolean model, which uses logic-based operators like AND and OR to decide whether documents match a user's query.

In this study, we evaluated a Boolean IR system built using the Whoosh library. The system was tested on a set of AI-related articles sourced from venturebeat.com. Our goal was to analyze how well the system performs across various types of queries, both simple and complex.

Methodology

Dataset

We worked with a dataset of 42 articles focused on artificial intelligence. These articles cover a variety of AI topics, including large language models, real-world applications, tech innovations, and societal impacts.

The Boolean IR System

Our system, built with Whoosh, goes through three main steps:

1. **Indexing:** Each document is indexed using a schema that includes fields like title, content, and path.
2. **Query Parsing:** User queries are parsed into Boolean logic expressions.
3. **Searching:** Documents are retrieved based on whether they satisfy the Boolean condition.

In this model, documents are treated as sets of terms. A document is considered relevant if it matches the entire Boolean query.

Evaluation Metrics

To measure the system's performance, we used standard IR metrics:

- **Precision:** How many of the retrieved documents are actually relevant?
 $\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$
- **Recall:** How many of the relevant documents were successfully retrieved?
 $\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$

- **F1-score:** A balance between precision and recall.

$$F1 = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

Queries Used

We tested the system using six queries:

Simple queries:

- "AI", "Nvidia", "open-source"

Complex queries:

- "AI AND Nvidia", "AI OR Google", "AI AND open-source"

Defining Relevance

To determine which documents were relevant for each query, we used a basic rule: a document was considered relevant if it contained the queried terms. While simple, this approach is appropriate for testing a Boolean model.

Results

Overall Performance

- **Average Precision:** 0.9762
- **Average Recall:** 1.0000
- **Average F1-score:** 0.9877

The system demonstrated excellent performance, retrieving all relevant documents and rarely including irrelevant ones.

Query-by-Query Breakdown :

Query	Precision	Recall	F1-score	Retrieved Docs	Relevant Docs
AI	1.0000	1.0000	1.0000	42	42
Nvidia	1.0000	1.0000	1.0000	13	13
open-source	0.9286	1.0000	0.9630	14	13
AI AND Nvidia	1.0000	1.0000	1.0000	13	13
AI OR Google	1.0000	1.0000	1.0000	42	42
AI AND open-source	0.9286	1.0000	0.9630	14	13
BANANA	0	0	0	0	1

Analysis

Strengths

- **Perfect Recall:** The system didn't miss any relevant documents for any query. (Except 'BANANA' one obviously.)
- **High Precision:** With nearly 98% precision on average, irrelevant results were minimal.
- **Stable Results:** Whether the query was simple or used Boolean operators, the system performed consistently well.

Key Takeaways

- **Boolean logic works:** Operators like AND and OR behaved as expected, allowing precise or broad searches.
- **Index quality matters:** Good normalization and consistent tokenization are critical.
- **Query phrasing is important:** Boolean systems demand clear and carefully written queries to avoid missing results.

Conclusion

Our evaluation shows that the Boolean IR system is highly effective for retrieving relevant documents when the query terms are well-defined. It achieved perfect recall and near-perfect precision across all tested cases.

Despite some of the inherent limitations of the Boolean model, it remains a powerful approach — especially for users who know exactly what they're looking for. Future enhancements could include:

- Adding NLP techniques to improve handling of term variations and synonyms
- Mixing Boolean logic with ranking models to improve flexibility

In conclusion, for well-scoped information needs, a Boolean IR system can still deliver precise and complete results.